AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I034]*

Team Name: *[TECH TITIANS]*

Domain: *[DOMAIN 1]*

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Presentation Outline

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Problem Statement

"AI-generated deepfakes can misuse a person's identity without their knowledge or consent. There is a need for a secure system to verify identity, detect manipulated content, and protect ownership and consent."

Problem Understanding

- [Who experiences this problem? Ordinary people, students, public figures, and anyone who receives a fake video and trusts it.]
- [What causes the problem? Free, easy AI tools that copy a face or voice from a few photos or seconds of audio, and no way to check consent.]
- [How does it affect users or stakeholders? Money lost to scams, damaged reputation, blackmail, and in rare cases loss of life.]

Existing Gaps & Challenges

- [Existing detection websites check only one media type, are costly and slow]
- [Research systems verify the publisher of a file, not the person shown in it. None asks for consent.]
- [Main challenge that remains unsolved is detection alone is unreliable, because detectors miss new kinds of fakes.]

Our Proposed Solution

- [Core idea or product A platform that checks a photo, video or audio file in three ways: detect, trace, ask.]
- [How the solution works at a high level AI gives a manipulation risk level, a blockchain lookup confirms whether the file is a registered original, and registered people shown in the file approve or decline.]
- [Main value delivered to users is four clear results (risk, origin, identity, consent), and a say over the use of your own face and voice.]

Technical Approach

[upload → fingerprint → origin check → AI detection → identity match → consent check → four results.]

[The backend receives the file, computes its fingerprint, queries the blockchain, runs the detector and face match, sends a consent request, and returns the results.]

Tech Stack

Frontend: [HTML,CSS,Javascript]

Backend: [Python]

Database: [Blockchain Database]

Innovation & Uniqueness

- [What is new or different? A consent check for the person depicted, signed and recorded so it can't be altered.]
- [Why is this approach suited to the problem? Origin and consent checks still work when AI detection is wrong.]
- [What is the key differentiator? Others ask "is this fake?" We also ask "did this person say yes?"]

Feasibility & Scalability

- **Feasibility:** [Built from existing pretrained models and open-source tools. Steps: web app, detector, consent database, hashing, then blockchain.]
- **Scalability:** [New detectors added as modules, then a forward-to-check bot on messaging apps, then platform partnerships.]
- **Risks:** [Wrong detection (show risk levels, human review), fake registration (identity proof at sign-up), works only inside our platform at first (grow through bot and partnerships).]

Cost for Product (Optional)

- Development or hardware cost: [0 (built by the team with free, open-source tools)]
- Hosting or maintenance cost: [about 1,500 to 3,000 a month for a small cloud server,]
- Estimated total: [20,000 to 40,000 for the first year]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: When a person who receives a suspicious video they upload a forwarded investment video and see it flagged as high risk with no origin record.
- Use case 2: When someone uploads a video showing a registered student. The student gets a request, declines, and the result shows “consent refused”.
- Use case 3: A college or company registers its official videos, so anyone can confirm a copy is the untouched original.

Impacts and Benefits

- **Benefit to users:** check before trusting, and control over their own identity.
- **Social and economic impact:** fewer scam losses, less spread of misinformation, protection from blackmail.
- **Measurable outcome:** detector accuracy on standard datasets, time taken per check, and number of registered users.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 16	Target [16.10]	[Peace, Justice and Strong Institutions]	[Protects people's rights and access to trustworthy information]
SDG 9	Target [9.5]	[Industry, Innovation and Infrastructure]	[Applies AI and blockchain research to a real digital safety problem]

Individual Member Contribution

Member	Contribution or responsibility
OVIYAA S	BLOCKCHAIN INTEGRATER
SHIVAMIHIT GOVINDARAJU	FRONTEND DEVELOPER
KAPIL KUMARAN K S	BACKEND DEVELOPER
MAADESH KUMAAR C	AI MULTIMODEL DEVELOPER

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